

Task 22: Cloud Service Providers Comparison

AIM

To explore major cloud service providers (AWS, Google Cloud, Microsoft Azure) and compare their core services related to compute, storage, databases, networking, and pricing models.

ALGORITHM

- Step 1: Identify the three major cloud providers: AWS, GCP, Azure.
- Step 2: List core compute services and compare features.
- Step 3: List core storage services and compare features.
- Step 4: List database services and compare options.
- Step 5: List networking services and compare capabilities.
- Step 6: Analyze the pay-as-you-use pricing model for each provider.
- Step 7: Summarize findings in a comparison table.

OUTPUT

Comparison of Cloud Service Providers:

1. COMPUTE SERVICES

- AWS: EC2 (Elastic Compute Cloud) - Virtual servers with 400+ instance types
- GCP: Compute Engine - VMs with per-second billing
- Azure: Virtual Machines - VMs integrated with Windows ecosystem

2. STORAGE SERVICES

- AWS: S3 (Simple Storage Service) - Object storage with 99.999999999% durability
- GCP: Cloud Storage - Unified object storage with multiple classes
- Azure: Blob Storage - Object storage with hot/cool/archive tiers

3. DATABASE SERVICES

- AWS: RDS (MySQL, PostgreSQL, etc.), DynamoDB (NoSQL), Aurora
- GCP: Cloud SQL, Cloud Spanner (globally distributed), Firestore
- Azure: Azure SQL Database, Cosmos DB (multi-model NoSQL)

4. NETWORKING SERVICES

- AWS: VPC, Route 53 (DNS), CloudFront (CDN), ELB

- GCP: VPC, Cloud DNS, Cloud CDN, Cloud Load Balancing
- Azure: Virtual Network, Azure DNS, Azure CDN, Load Balancer

5. SERVERLESS COMPUTE

- AWS: Lambda - Pay per invocation, 15-min max timeout
- GCP: Cloud Functions - Event-driven, pay per invocation
- Azure: Azure Functions - Event-driven, consumption plan

6. PRICING MODEL (Pay-As-You-Use)

All three providers follow the pay-as-you-go model:

- No upfront costs or termination fees
- Billed per second/minute/hour of resource usage
- Free tier available (AWS: 12 months, GCP: \$300 credit, Azure: \$200 credit)
- Reserved instances offer 40-75% discounts for 1-3 year commitments
- Spot/Preemptible instances offer up to 90% savings for interruptible workloads

7. MARKET SHARE (2024)

- AWS: ~31% (Market leader, largest service catalog)
- Azure: ~25% (Strong enterprise/hybrid cloud)
- GCP: ~11% (Strong in data/AI/ML services)

RESULT

Successfully explored and compared AWS, Google Cloud, and Microsoft Azure across compute, storage, database, networking, and pricing. Each provider follows the pay-as-you-use model with free tiers and discount options for committed usage.